Alaa Ali Mousa

Flutter Developer — Cross-Platform Mobile Engineer — AI-Integrated Applications

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Summary

Flutter Developer with hands-on experience building cross-platform mobile applications using Dart, Flutter, and Firebase, plus full-stack experience integrating FastAPI backends and AI/deep-learning models into production-style mobile systems. Led the mobile and system-integration work on IntraVision, a full-stack AI medical imaging application, from architecture through deployment. Skilled in Clean Architecture, state management (Provider, Riverpod, BLOC), RESTful API integration, and real-time data, with a strong focus on performance, UX, and end-to-end system design.

Education

Delta Technology University Sept 2022 – June 2026

B.Sc. in Information Technology and Software

Overall Grade: Very Good — Achieved Distinction (Excellence Appreciation) in three academic years

Skills

- **Languages & Frameworks:** Flutter, Dart, Python, FastAPI, SQL, JSON
- **Mobile Architecture:** Clean Architecture, BLOC/Cubit, Provider, Riverpod, setState, Dependency Injection (get_it)
- **Backend & Data:** Firebase (Auth, Firestore, FCM), MongoDB, REST API Design, JWT Authentication, OOP
- **AI & Integration:** AI/Deep Learning Model Integration, Third-Party API Integration, Data Processing
- **UI/UX:** Component-Based UI, Responsive Design, Light/Dark Mode, Localization (Arabic/English)
- **Tools & Practices:** Git, GitHub, Postman, Android Studio, Visual Studio Code, Agile Development

Experience

IntraVision — AI-Powered Brain Tumor Segmentation System 2025 – 2026

Graduation Project · Flutter, FastAPI, Python, AI Integration, MongoDB

- Developed a full-stack medical system facilitating MRI processing, automated AI tumor segmentation, and report generation.
- Constructed a high-performance FastAPI backend with MongoDB Atlas, handling multi-source data workflows and 32+ API endpoints.
- Integrated deep learning inference pipelines (CycleGAN and nnU-Net v2) to process volumetric image metrics and medical data.

Freelance Flutter Developer 2025

Nafezly · Remote

- Developed responsive Flutter UI screens for Android and iOS applications.
- Implemented reusable UI components and organized, maintainable code structures.
- Worked on Flutter interface development with potential integration of APIs, Firebase, and authentication based on project requirements.

Freelance Flutter Developer 2025

Kafiil · Remote

- Developed a responsive portfolio application using Flutter, featuring profile information, technical skills, project listings, and contact screens.
- Implemented reusable UI components, screen navigation, form validation, and locally stored mock data.
- Structured the application for maintainability and future customization across different mobile screen sizes.

Projects

Flutter Developer Trainee — Digital Egypt Pioneers Initiative (DEPI) 2025 – 2026

Ministry of Communications and Information Technology (MCIT), Egypt

- Completed an intensive, project-based Flutter Developer training program covering cross-platform mobile development and modern software engineering practices.
- Built and shipped multiple Flutter applications applying Firebase (Authentication, Firestore), RESTful API integration, and state management with Provider and Riverpod.
- Collaborated in team-based projects, strengthening communication, problem-solving, and time-management skills in an Agile workflow.

Intra Vision 2025 – 2026

AI-Powered Brain Tumor Segmentation System · Graduation Project, Full-Stack Developer (Flutter, FastAPI, AI Integration)

- Built a full-stack medical imaging platform enabling doctors to upload MRI scans, auto-generate contrast-enhanced images via deep learning, segment brain tumors, and produce AI-written medical reports from a mobile app.
- Developed the Flutter Android app with Clean Architecture and BLOC/Cubit state management, including biometric login, secure storage, and push notifications (Firebase Cloud Messaging).
- Integrated a CycleGAN model to synthesize contrast-enhanced (T1CE) MRI from native T1 scans and an nnU-Net v2 pipeline for automated tumor segmentation, returning confidence score, volume, and diameter metrics.
- Built a FastAPI backend with MongoDB Atlas, JWT authentication, 2FA email verification, and 32+ REST endpoints; generated AI-written PDF reports using Google Gemini and WeasyPrint.
- Implemented interactive 3D brain/tumor visualization (trimesh, model_viewer plus) and a QR-code-based patient portal for real-time case tracking.
- Diagnosed and resolved critical production issues, including native crashes on large NIfTI file handling and out-of-memory errors on 150MB+ downloads, by redesigning the file I/O pipeline.

Flower App (Flutter)
Built an e-commerce app with cart, favourites, and product filtering, featuring smooth animations and a responsive UI.

News App (Flutter)
Built a news app fetching live global news via a REST API, with smooth navigation and caching for better performance.

Chat App (Flutter + Firebase)
Built a real-time chat app using Firebase Authentication and Firestore, with message timestamps, chat rooms, and user login.

Weather App (Flutter)
Built a weather app using the OpenWeatherMap API with a dynamic UI that adapts to temperature and conditions.

Smart Irrigation System (IoT)
Built an automated irrigation system with Arduino, water pumps, and moisture sensors for real-time monitoring and control.

Courses & Certifications

- **Flutter & Dart:** Creativa Menof (2025) – Cross-platform mobile development and state management fundamentals.
- **Flutter & Dart:** Easy Learn Academy (2025) – Real-world app development with Firebase integration and REST APIs.
- **Artificial Intelligence:** Huawei ICT Academy (2025) – Machine learning fundamentals and AI workflow applications.
- **Data Science:** Cisco (2024) – Data analysis, processing, and statistical fundamentals.
- **Internet of Things (IoT):** Cisco (2024) – Sensors, controllers, and connected systems for automation.

Languages

Arabic (Native) · English (Intermediate)